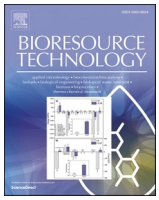
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Alleviation of ammonia inhibition via nano-bubble water supplementation during anaerobic digestion of ammonia-rich swine manure: Buffering capacity promotion and methane production enhancement

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HIGHLIGHTS

- Ammonia inhibition was alleviated under nano-bubble water (NBW) supplementation.
- Extracellular hydrolases and coenzyme F_{420} were enhanced with NBW supplementation.
- NBW supplementation promoted buffering capacity under ammonia inhibition.
- CH_4 production was enhanced by 12.3–38.7% by NBW supplementation.
- NBW addition shortened the lag phase and increased the maximum CH_4 production rate.

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ABSTRACT

Anaerobic digestion (AD) of ammonia-rich swine manure (SM) with nano-bubble water (NBW) supplementation was studied in this work with the expectation of ammonia inhibition alleviation, buffering capacity promotion, and methane production enhancement. Results indicated that cumulative methane yield was elevated by 12.3–38.7% in NBW groups. Besides, the reduced methane production rate and elongated lag phase under ammonia inhibition were increased and shortened by NBW supplementation, respectively. The rapid increase of total alkalinity (TA) and partial alkalinity (PA) could be observed with NBW supplementation, as well as the rapid decline of VFA/TA, thus improved buffering capacity and alleviated ammonia inhibition. Moreover, higher level of extracellular hydrolases and coenzyme F_{420} could be detected in NBW groups. In conclusion, NBW with higher mobility and zeta potential (absolute value) could be a promising strategy for the alleviation of ammonia suppression during the AD of SM.

1. Introduction

Over 80 million tons of livestock manure are produced in Japan annually, among which the output of swine manure (SM) exceeds one-quarter of the total (around 21 million tons) (MAFF, 2019). To reduce the pollution in surrounding environment and increase the efficiency of biomass resources utilization, these enormous amounts of SM should be treated scientifically and effectively. Composting is widely applied to treat livestock manure due to its many advantages, such as increasing fertility and stabilizing organics (Yuan et al., 2017). However, composting has shortcomings like excessive carbon dioxide emissions and nutrient loss, which restrict its wider application in practice (Fukumoto et al., 2003). Compare to composting, anaerobic digestion (AD) shows various superiorities in solid reduction and energy production when treating livestock manure (Li et al., 2020). However, ammonia inhibition occurs easily during AD processes due to the high nitrogen content of SM, which is not good for the progress of AD (Huang et al., 2019). If ammonia inhibition cannot be effectively alleviated, it will affect the AD system and reduce methane yield (Tian et al., 2019).

Up to now, various methods have been applied in previous studies to

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alleviate ammonia inhibition such as ammonia stripping (Yin et al., 2019), membrane (Shi et al., 2020), microbial acclimation (Yan et al., 2019) and struvite precipitation (Lorick et al., 2020). However, these technologies still have disadvantages, such as complexity of operation, energy consumption and secondary pollution. New strategies with easier operation processes and less chemical addition are desired for efficient alleviation of ammonia inhibition. As known, water plays a prominent role in the buffering and material transport for all the substrate and reactants (Kothari et al., 2014; Bollon et al., 2013). In this field, most of the previous reports focused on the impact of water content on the AD process, namely, the impact of different total solids (TS) on AD process (Indren et al., 2020; Latifi et al., 2019). However, few studies focused on the influence of water characteristics and types on the AD process under ammonia inhibition. It's still unclear whether changing the types and characteristics of water would bring about improvement in AD performance under ammonia inhibition.

Nano-bubble water (NBW) has some unique characteristics such as high air solubility, improved mass transferability and generation of free radicals (Liu et al., 2019, 2013; Temesgen et al., 2017). NBW has been widely used in medicine (Hayashi et al., 2019), functional food (Guo et al., 2020), agriculture (Zhou et al., 2019), industry (Kim et al., 2020), and environmental engineering (Yu et al., 2019). Most recently, NBW was found to be able to promote methane production during the AD process as reported by many studies (Fan et al., 2020a; Yang et al., 2020a). However, the above-mentioned promoting effect by NBW was achieved when no inhibition occurred. Considering the unique properties of NBW, it would be of great promise in alleviating ammonia inhibition and enhancing methane production in the AD of ammonia-rich SM, without secondary pollution and exceeding energy consumption.

Therefore, the purpose of this work was to elucidate the effect of NBW supplementation on the AD of ammonia-rich SM occurred varying degrees of ammonia inhibition, aiming at applying a new approach to alleviate ammonia inhibition and enhance methane production without using conventional methods. This research was the first attempt of applying NBW in the AD process of ammonia-rich SM. Results from this work are targeting a novel method to alleviate the inhibitory effect from ammonia, which often occurs during the AD process in ammonia-rich SM.

2. Materials and methods

2.1. AD tests

2.1.1. SM and inoculum

The fresh SM was directly collected from the medium-sized swine company in Tsukuba, Ibaraki Prefecture, Japan. Before stored at 4 °C for subsequent tests, this fresh SM was mixed thoroughly to obtain the representative SM sample. Digested sludge used for acclimatization of inoculum was directly taken from a sewage treatment plant in Ibaraki Prefecture, Japan in this work. The storage of directly collected digested sludge and the process of acclimatization of inoculum were conducted based on the previous report (Fan et al., 2020b). The main characteristics of the fresh SM and inoculum were shown in Table 1.

2.1.2. AD trials

In this work, different amounts of NH_4Cl were added to the reactors before the AD experiment to simulate different levels of ammonia inhibition. Eight groups of reactors were designed: NH_4Cl (g) in $R(0)_{Control}$/$R(0)_{NBW}$; $R(1)_{Control}$ /$R(1)_{NBW}$; $R(3)_{Control}$ /$R(3)_{NBW}$; $R(5)_{Control}$ /$R(5)_{NBW}$ were addition to 0, 1 g, 3 g and 5 g, respectively. Each Scott Duran bottle (working volume 200 mL) was loaded with SM and inoculum, and the feed/microorganisms (VS basis) ratio was kept about 1:1. At the beginning of the tests, 100 mL NBW or deionized water (DW) were added into R_{NBW} and $R_{Control}$, respectively. To avoid the introduction of chemicals and to reduce changing the structure of the microbial communities, pH adjustment was not performed during the whole AD process. After being loaded with inoculum, SM and NBW or DW, all reactors were sealed. N_2 was then introduced for 3 min in all groups of reactors to ensure an initial anaerobic environment. Afterwards, all these groups of reactors were placed in an incubator at 35 ± 2 °C for about 42 days according to the preliminary experiments. All AD experiments were conducted in triplicate.

Table 1
Characteristics of raw swine manure and inoculum used in this study.

Parameters	Unit	Inoculum	Raw swine manure
Total solids (TS)	%	1.29 ± 0.09	32.46 ± 0.41
Volatile solids (VS)	%	0.95 ± 0.06	27.46 ± 0.29
Total ammonia nitrogen (TAN)	mg/g	0.38 ± 0.02	3.53 ± 0.17
Soluble total carbon (STC)	mg/g	0.87 ± 0.02	39.54 ± 0.53
Soluble inorganic carbon (SIC)	mg/g	0.44 ± 0.01	3.45 ± 0.03
Soluble total organic carbon (STOC)	mg/g	0.43 ± 0.01	36.09 ± 0.49
Volatile fatty acids (VFAs)	mg/g	0.14 ± 0.02	35.31 ± 1.82
Total alkalinity (TA, as CaCO3)	mg/g	2.34 ± 0.04	17.27 ± 0.25
Soluble proteins	mg/g	0.31 ± 0.02	11.07 ± 0.33
Soluble carbohydrates	mg/g	0.11 ± 0.01	6.30 ± 0.18
C	%/TS	ND	43.45
H	%/TS	ND	6.70
N	%/TS	ND	3.77
S	%/TS	ND	0.79
C/N ratio	–	ND	11.53
pH	–	7.17 ± 0.04	6.26 ± 0.05

ND (Not Determined).
Data in the table are expressed as Means ± SD.

2.1.3. Analytical methods

In this work, the methane production from SM under different levels of ammonia inhibition was compared and analyzed. Total ammonia nitrogen (TAN), volatile solids (VS), and TS were measured by using the standard method (APHA, 2012). The methane content, pH value, the content of C, H, N, and S, soluble total organic carbon (STOC), volatile fatty acids (VFAs), free ammonia nitrogen (FAN), intermediate alkalinity (IA), total alkalinity (TA), and partial alkalinity (PA) were analyzed according to a previous study (Fan et al., 2021). Soluble proteins and soluble carbohydrates were analyzed by the Lowry method and the phenol–sulfuric method, respectively. The relative activities of three extracellular hydrolases, including acid phosphatase (ACP), alkaline phosphatase (ALP), and α-glucosidase were determined on day 6 based on a previous literature (Goel et al., 1998). In order to study the growth status of methanogens, coenzyme F_{420} was measured according to the previous reports at the end of the test (Dong et al., 2019; Delafontaine et al., 1979).

2.2. NBW preparation and characterization

According to the previous work (Yang et al., 2019), Air-NBW was produced by applying a nano-bubble generation machine (HACK FB11, JAPAN). During the NBW production process, the preparation time was controlled at 20 min, and the pressure in NBW generator was kept at around 0.25 MPa. After that, NBW obtained from this nano-bubble generator was used throughout the whole AD process. NBW was characterized based on the previously reported methods (Wang et al., 2019, Guo et al., 2019).

2.3. Calculation methods

Relative enzyme activity (%) was determined by Eq. (1)

$$\text{Relative enzyme activity } (\%) = \frac{V_{NBW}}{V_{DW}} \times 100 \quad (1)$$

where V_{NBW} is the absorbance value at a specific wavelength of R_{NBW}, V_{DW} is the absorbance value of $R_{Control}$ at the same wavelength as R_{NBW}.

2.4. Statistical analysis

Since NBW characteristics and the AD tests under different levels of ammonia inhibition were conducted in triplicate, mean value ± standard deviation (SD) were used for results and discussion. Statistical analysis was done using one-way analysis of variance (ANOVA) followed by Tukey's post-hoc test using Microsoft Excel 2019. The different letters (a, b, c, …) indicated the statistical significance ($p < 0.05$). Origin 9.0 (Origin Lab) was also applied for the analysis and draw.

3. Results and discussion

3.1. TAN, FAN, and soluble organics

Fig. 1a showed the changes of TAN with the supplementation of DW or NBW. AD process would be suppressed when TAN content exceeds 3,000 mg/L (Bi et al., 2020). The difference in TAN content of R_{NBW} and the corresponding control groups was not statistically significant ($p > 0.05$). Since $R(0)_{NBW}$ and $R(0)_{Control}$ did not exceed 1,000 mg/L, ammonia inhibition did not occur in these two groups. Severe ammonia inhibition also did not appear in $R(1)_{Control}$ and $R(1)_{NBW}$ because the TAN content of these two groups did not exceed 3,000 mg/L. $R(3)_{NBW}$ and the corresponding control group, however, showed ammonia inhibition due to the TAN content higher than 3,500 mg/L. TAN content exceeding 6,000 mg/L appeared in $R(5)_{Control}$ and the corresponding control group, indicating that severe ammonia inhibition occurred in these two groups. The changes of FAN were shown in Fig. 1b. Obviously, FAN inhibition did not appear in $R(0)_{Control}$ and $R(0)_{NBW}$ because the FAN content of these two groups was <30 mg/L. According to a previous report, during AD process, the threshold for FAN inhibition was about 80 mg/L (De Baere et al., 1984). As FAN content in $R(3)_{NBW}$ and the corresponding control group exceeded 90 mg/L and was significantly

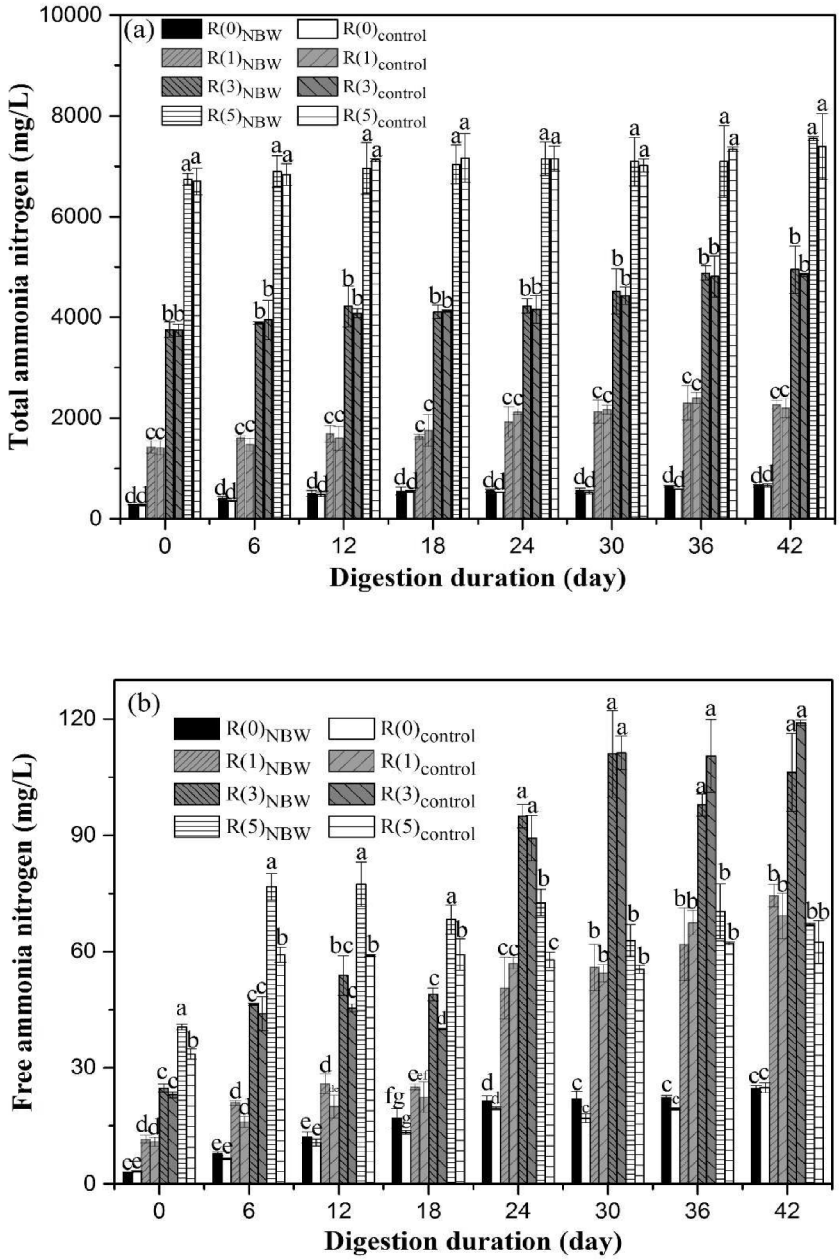


Fig. 1. Change of total ammonia nitrogen and free ammonia nitrogen variation; (a) Total ammonia nitrogen variation. (b) Free ammonia nitrogen variation. One-way analysis of variance (ANOVA) followed by Tukey's post-hoc test was applied for the statistical analysis of total ammonia nitrogen and free ammonia nitrogen. The different letters (a, b, c, …) indicate significant difference at the 5% level ($p < 0.05$).

higher than the other groups of reactors ($p < 0.05$) after 24 days, FAN inhibition occurred in these two groups. Although the TAN content of R(5)$_{NBW}$ and R(5)$_{Control}$ was higher than 6,000 mg/L, the FAN content of these two groups did not exceed 90 mg/L, which was caused by the lower pH value. As for R(1)$_{NBW}$ and R(1)$_{Control}$, FAN inhibition might occur in the late AD process due to FAN content above 60 mg/L.

The VFAs concentrations in all groups were shown in Fig. 2a and Fig. 2b. R(0)$_{NBW}$ and R(0)$_{Control}$ increased firstly and then decreased rapidly and VFAs accumulation did not occur during the whole AD processes. VFAs accumulation appeared in the early stage of AD in R(1)$_{NBW}$ and R(1)$_{Control}$. On day 12, day 18, and day 24, R(1)$_{NBW}$ was lower than R(1)$_{Control}$ ($p < 0.05$). Combining with the results of methane production (Fig. 4), the lower VFAs concentrations in R(1)$_{NBW}$ may be due to higher consumption than that in R(1)$_{Control}$. That was, NBW supplementation might promote the consumption of VFAs when ammonia suppression was not severe. The VFAs of R(3)$_{NBW}$ and the corresponding control group accumulated in the hydrolysis-acidification stage and then decreased in the methanation process. However, when the methanogenesis process was completely suppressed, the VFAs of R(5)$_{Control}$ and R(5)$_{NBW}$ could not be consumed during the AD processes. Therefore, NBW supplementation could not change the AD process failure when severe ammonia inhibition occurred.

In the beginning, the concentrations of soluble proteins/carbohydrates in all groups were almost at the same level (Fig. 2c). The soluble proteins were first increased and then rapidly consumed in R(0)$_{NBW}$ and R(0)$_{Control}$. Although the soluble proteins in R(3)$_{Control}$ and the corresponding control group also decreased rapidly in the late stage of AD, these two groups were higher than that of R(0)$_{NBW}$ and R(0)$_{Control}$. This phenomenon indicated that the higher TAN content was not conducive to the consumption of soluble protein. As for R(5)$_{Control}$ and R(5)$_{NBW}$, the consumption of soluble protein was suppressed since the inhibition of AD process by high TAN content. The soluble proteins of R(0)$_{NBW}$, R(1)$_{NBW}$, and R(3)$_{NBW}$ were higher than the corresponding R(0)$_{Control}$, R(1)$_{Control}$, and R(3)$_{Control}$ on the 6th day. In terms of soluble carbohydrates, R(5)$_{Control}$ and R(5)$_{NBW}$ were also not consumed in large amounts during the AD process due to severe ammonia inhibition. Other groups, however, were gradually consumed as AD progressed, among which R(3)$_{Control}$ and R(3)$_{NBW}$ accumulated before being consumed. This was consistent with the performance of VFAs (Fig. 2a and Fig. 2b).

Fig. 2d illustrated the changes in soluble TOC throughout the 42 days' AD process. Similarly, the soluble TOC contents of R(5)$_{Control}$ and R(5)$_{NBW}$ experienced an increasing firstly and was not consumed largely during the AD process. Other groups, however, firstly increased and then gradually consumed. It was worth noting that the final contents of R(3)

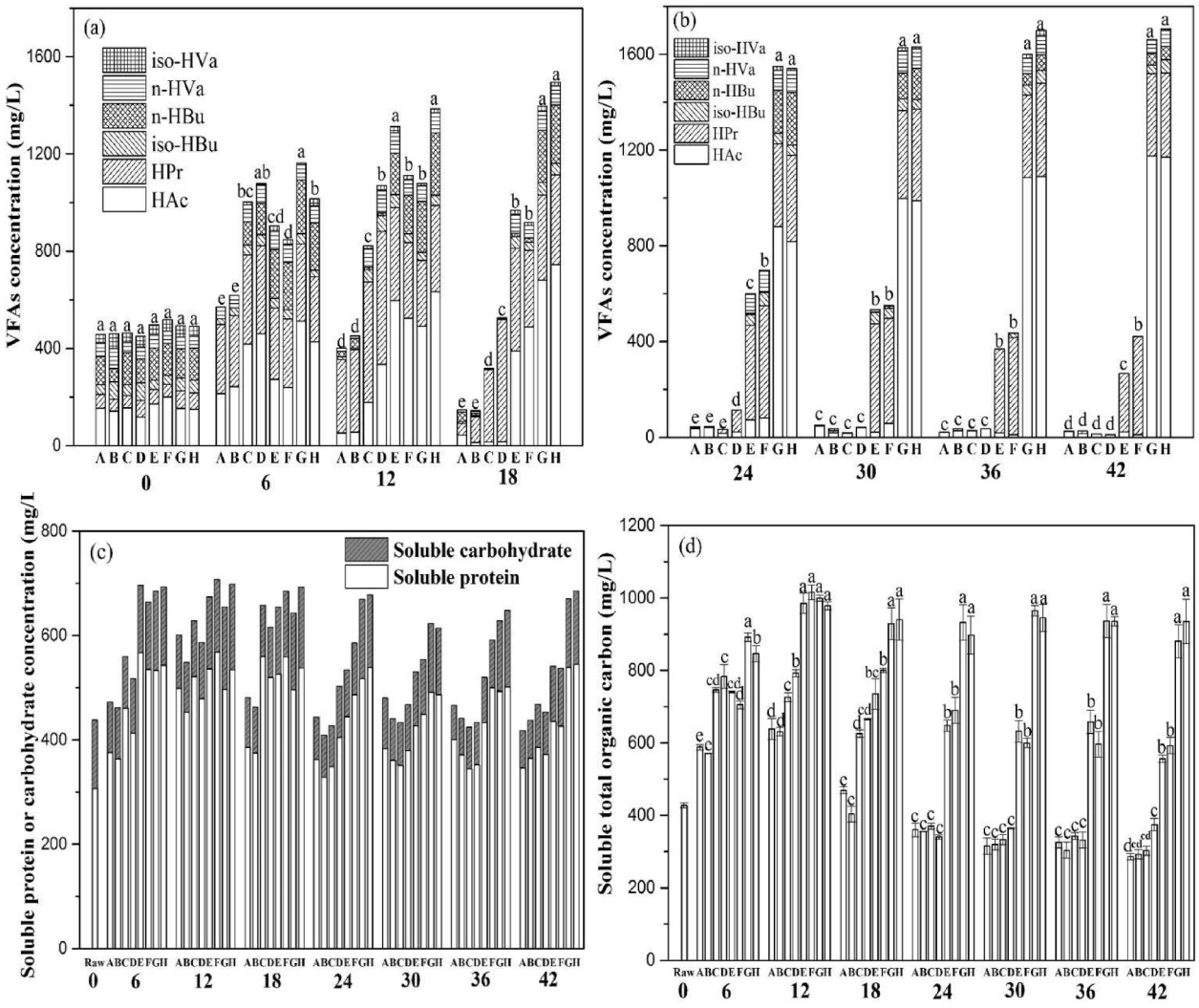


Fig. 2. VFAs concentration, soluble protein/carbohydrate, and soluble total organic carbon; (a) VFAs on day 0, day 6, day 12, and day 18. (b) VFAs on day 24, day 30, day 36, and day 42. (c) Soluble protein/carbohydrate. (d) Soluble total organic carbon. The alphabets under X-axis areas: A- R(0)$_{NBW}$, B- R(0)$_{Control}$, C- R(1)$_{NBW}$, D- R(1)$_{Control}$, E- R(3)$_{NBW}$, F- R(3)$_{Control}$, G- R(5)$_{NBW}$, H- R(5)$_{Control}$. One-way analysis of variance (ANOVA) followed by Tukey's post-hoc test was applied for the statistical analysis of VFAs and soluble total organic carbon. The different letters (a, b, c, …) indicate significant difference at the 5% level ($p < 0.05$).

were higher than those of R(0) and R(1) ($p < 0.05$). A similar phenomenon also appeared in the performance of VFAs (Fig. 2b). These phenomena occurred because ammonia inhibition was detrimental to the consumption of organics, resulting in a higher final soluble TOC and VFAs contents in $R(3)_{NBW}$ and the corresponding control group. In addition, the final VFAs contents of $R(3)_{NBW}$ were lower than that of the corresponding control group ($p < 0.05$), indicating that NBW supplementation may promote the consumption of VFAs, which could also explain the significantly higher cumulative methane production in $R(3)_{NBW}$ compared to that of $R(3)_{Control}$ (Fig. 4).

3.2. Buffering capacity of the AD process

Since TA, IA, and PA are indicators for evaluating the stability during the AD process (Sun et al., 2016), these indicators were analyzed in this study. The changes in TA content under different levels of ammonia inhibition were shown in Fig. 3a. The TA of all groups increased to varying degrees, which was consistent with the change of pH value (Fig. 3e). The TA content of $R(0)_{NBW}$ and $R(1)_{NBW}$ were higher than the corresponding $R(0)_{Control}$ and $R(1)_{Control}$, respectively on the 6th day ($p < 0.05$). In addition, on the 12th day, $R(1)_{NBW}$ and $R(3)_{NBW}$ were higher than the corresponding $R(1)_{Control}$ and $R(3)_{Control}$, respectively ($p < 0.05$). These phenomena showed that during the early stages of AD process, NBW might benefit the increase in TA under ammonia inhibition. However, $R(5)_{NBW}$ and $R(5)_{Control}$ exhibited the lowest TA content, which was due to severe ammonia inhibition. Therefore, when severe ammonia inhibition occurred, NBW supplementation could not effectively promote the buffering capacity of the AD system.

PA content also experienced an upward trend in the early stage of AD (Fig. 3b). The relatively lowest PA content appeared in $R(5)_{Control}$ and $R(5)_{NBW}$, which indicated that NBW supplementation could not promote the buffering capacity of AD system under severe ammonia inhibition. Comparison with $R(0)_{Control}$ and $R(1)_{Control}$, on the 6th day, $R(0)_{NBW}$ and $R(1)_{NBW}$ showed a higher PA ($p < 0.05$), indicating that NBW was beneficial to the increase of PA and thus promoted the buffering capacity. Since IA/PA can be applied as a parameter to monitor the stability during the AD process (Yirong et al., 2017), this work also focused on the IA/PA ratio. All groups were reduced, except for $R(5)_{NBW}$ and $R(5)_{Control}$ (Fig. 3c). As for $R(5)_{Control}$ and $R(5)_{NBW}$, the IA/PA ratio did not decrease significantly, which confirmed that the buffering capacity of these two groups was poor.

Fig. 3d illustrated the changes in VFA/TA ratio. In the beginning, the VFA/TA ratio in R_{NBW} and the corresponding $R_{Control}$ were almost at the same level. On the 6th day, the VFA/TA ratio in $R(0)_{NBW}$ gradually decreased, but other groups increased, indicating that $R(0)_{NBW}$ had the best-buffering capacity performance. The VFA/TA ratio of all groups gradually decreases with the AD process progressed except for $R(5)_{NBW}$ and $R(5)_{Control}$. The VFA/TA of $R(0)_{NBW}$ and $R(1)_{NBW}$ were higher than the corresponding $R(0)_{Control}$ and $R(1)_{Control}$, respectively on the 6th and 12th day, and the difference was significant ($p < 0.05$). In addition, $R(1)_{NBW}$ and $R(3)_{NBW}$ were higher than the corresponding $R(1)_{Control}$ and $R(3)_{Control}$, respectively on the 24th day, and the difference was also significant ($p < 0.05$). These similar phenomena further showed that NBW could promote the buffering capacity during the AD process. However, NBW supplementation could not promote the buffering capacity under sever ammonia inhibition. As an important parameter, pH is often applied for evaluating the buffering ability (Veluchamy et al., 2019). All groups of reactors gradually increased, except for $R(5)_{NBW}$ and $R(5)_{Control}$ (Fig. 3e). This also showed that the AD processes of $R(5)_{NBW}$ and $R(5)_{Control}$ failed due to severe ammonia inhibition.

The above results indicated that NBW supplementation could promote buffering capacity and alleviate slight ammonia inhibition. However, when TAN content was too high, the AD process was completely suppressed, and NBW supplementation could not change the failure of the AD process. This was because the supplementation of NBW to promote AD has a threshold in the case of ammonia inhibition. When the content of TAN and FAN was below the threshold, the supplementation of NBW could promote the AD process since the ammonia inhibition was not serious. However, when the content of TAN and FAN exceeded the threshold, the promotion effect of NBW was not enough to alleviate severe ammonia inhibition. Regarding the threshold at which NBW could significantly promote the AD process under ammonia inhibition, further research is needed.

3.3. Methane production

As shown in Fig. 4, when severe ammonia inhibition did not occur, the methane production from NBW groups were higher than that of control groups ($p < 0.05$). When TAN content exceeded 7,000 mg/L, the AD process of $R(5)_{NBW}$ and $R(5)_{Control}$ was inhibited. For the methane production, $R(0)_{NBW}$, $R(1)_{NBW}$ and $R(3)_{NBW}$ obtained 218, 185 and 107 mL/g-VS in the average, respectively, increasing by 15.3%, 12.3%,

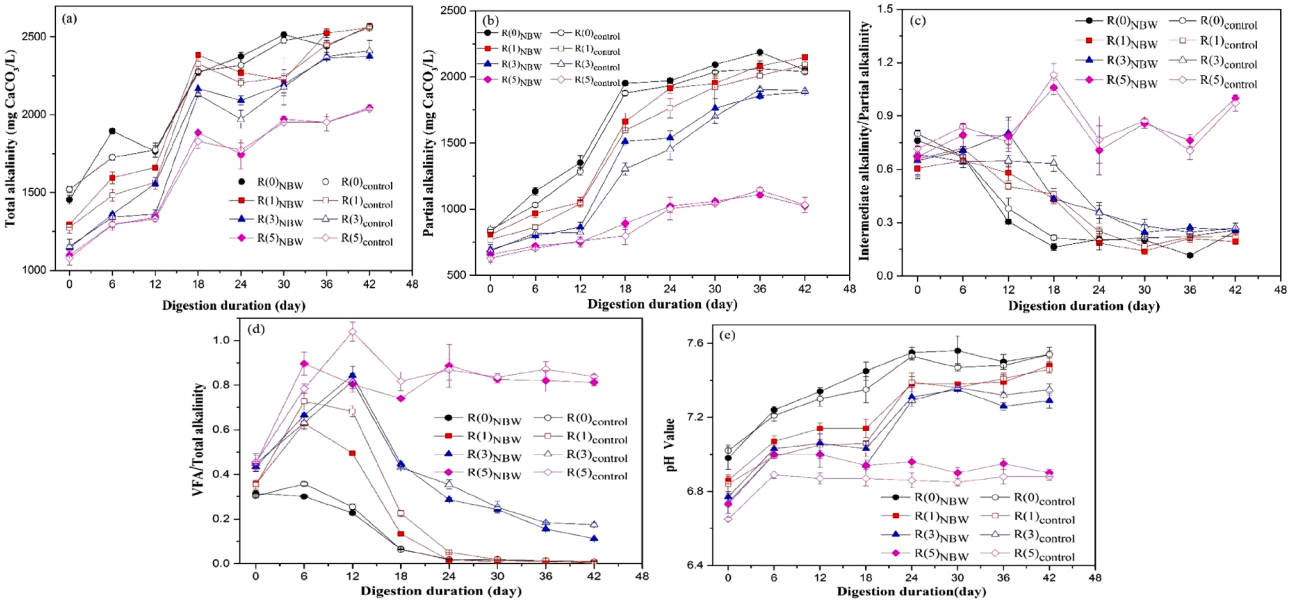


Fig. 3. (a) Total alkalinity. (b) Partial alkalinity. (c) Intermediate alkalinity/Partial alkalinity. (d) VFA/Total alkalinity. (e) pH value.

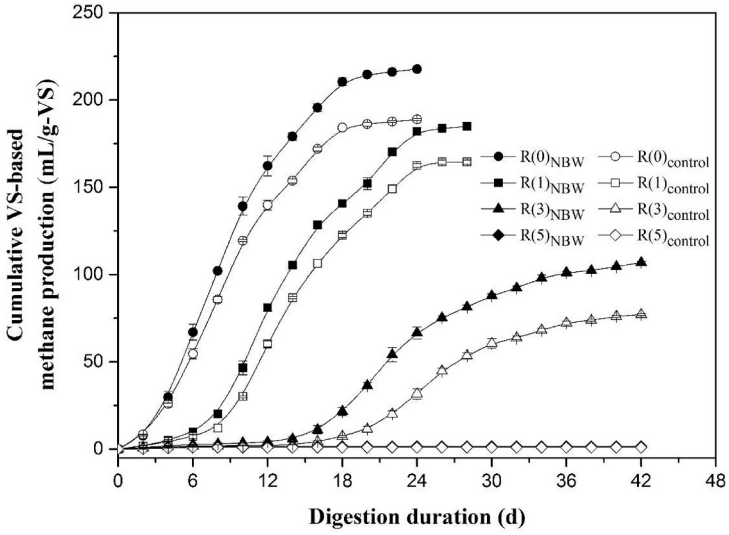


Fig. 4. Cumulative VS-based methane production.

38.7% in comparison to the corresponding control groups, respectively. In these eight groups, the maximum production of methane appeared in $R(0)_{NBW}$. With the increasing of TAN, the yield of methane decreased.

To better analyze the methane yield performance, the modified Gompertz model was used based on the previous research (Wang et al., 2019). The results were showed in Table 2. The $R(1)_{NBW}$ and $R(3)_{NBW}$ showed a relatively shorter lag phase at a significant difference ($p < 0.05$) compared to those control groups. This result indicated that NBW significantly shorten the lag phase under ammonia inhibition. Obviously, the methane production performance of R_{NBW} was better than the corresponding $R_{Control}$, which could be confirmed by the higher *P*max value in R_{NBW}. Moreover, in terms of R_{max}, $R_{Control}$ was also significantly less than the R_{NBW} ($p < 0.05$). These parameters were significantly promoted by NBW supplementation indicated that NBW supplementation enhanced the maximum methane production rate under ammonia suppression.

3.4. Related enzyme

During the AD process, hydrolysis is an essential stage because hydrolysis is considered the rate-limiting step (Yan et al., 2020). To reveal the application effect of NBW supplementation on the hydrolysis period under different levels of ammonia inhibition, this work evaluated three important extracellular hydrolases (ACP, ALP, and α-glucosidase). The relative activities of the three extracellular hydrolases in R_{NBW} were higher than the corresponding $R_{Control}$, indicating that NBW supplementation could increase the activity of extracellular hydrolases to accelerate the hydrolysis process under ammonia inhibition (Fig. 5a). For the methanogenesis stage, the activity of methanogens has an important influence on this stage (Srisowmeya et al., 2020). Coenzyme F_{420} is a common coenzyme for methanogenic bacteria and can be used to determine the potential methanogenic bacterial activity in mixed microbial communities (Delafontaine et al., 1979). Clearly, the activity of coenzyme F_{420} with NBW supplementation was enhanced relative to the corresponding $R_{Control}$ (Fig. 5b). When the AD process progressed to the methanogenesis stage, the relatively high activity of coenzyme F_{420} and the increased methane production in R_{NBW} showed that the activity of methanogens was enhanced due to NBW supplementation under ammonia inhibition. Therefore, under ammonia inhibition, NBW could not only accelerate the hydrolysis process but also increase the activity of methanogens and further improve the methane production.

3.5. Possible underlying mechanisms and application prospects

This work confirmed that NBW supplementation could enhance methane production and relieve ammonia inhibition in ammonia-rich SM. NBW contains hundreds of millions of nano-bubbles (NBs) with a diameter of about a few hundred nm (Table 3), which are not available in DW. Moreover, the zeta potential of DW was lower than NBW for the absolute value, and the difference was significant ($p < 0.05$). The colloid has a stable system and avoids aggregation when zeta potential is high (Xiao et al., 2020). Therefore, NBW was more inclined to avoid the aggregation and maintain a stable system than DW, indicating that the stability of R_{NBW} may be better relative to $R_{Control}$. In addition, NBW has relatively longer spin–spin relaxation times (T_2) and higher water molecule mobility compared to ordinary water (Liu et al., 2013; Fan et al., 2020a). Therefore, NBW with high water molecule mobility and high stability promoted the AD process under ammonia inhibition. In the hydrolysis stage, the enhanced hydrolase activity due to the supplementation of NBW promoted the hydrolysis process. Moreover, NBW with higher mobility could improve nutrient mass transfer efficiency from liquid to bacterial cells (Wang et al., 2019). This was beneficial for improving the activity of microorganisms. Previous reports confirmed that the activity of Methanobacterium, Methanomassiliicoccus, Methanospirillum, and Methanosaeta was enhanced due to the supplementation of NBW (Wang et al., 2020). This showed that the supplementation of NBW enhanced the activity of hydrogenotrophic methanogens (HM) and acetoclastic methanogens (AM). 70% of

Table 2
Kinetic analysis of cumulative CH_4 production from AD of SM with and without adding NBW under different ammonia inhibition.

Model parameters	$R(0)_{NBW}$	$R(0)_{Control}$	$R(1)_{NBW}$	$R(1)_{Control}$	$R(3)_{NBW}$	$R(3)_{Control}$
R^2	0.9990	0.9987	0.9974	0.9978	0.9977	0.9974
λ (d)	2.60 ± 0.14 ^{e}	2.69 ± 0.16 ^{e}	6.56 ± 0.26 ^{d}	7.49 ± 0.24^{c}	14.64 ± 0.27^{b}	17.89 ± 0.28 ^{a}
Pmax (mL CH_4/g-VS)	223.05 ± 1.94 ^{a}	195.00 ± 1.96^{b}	193.46 ± 3.64^{b}	174.33 ± 3.23^{c}	107.22 ± 1.42 ^{d}	79.38 ± 1.31 ^{e}
Rmax (mL CH_4/g-VS·d)	18.98 ± 0.47 ^{a}	16.32 ± 0.46^{b}	13.97 ± 0.58^{c}	12.79 ± 0.40 ^{d}	6.98 ± 0.25 ^{e}	5.38 ± 0.21^{f}

Data in the table are expressed as Means ± SD, and the different letters (a, b, c, …) indicate significant difference at the 5% level ($p < 0.05$).

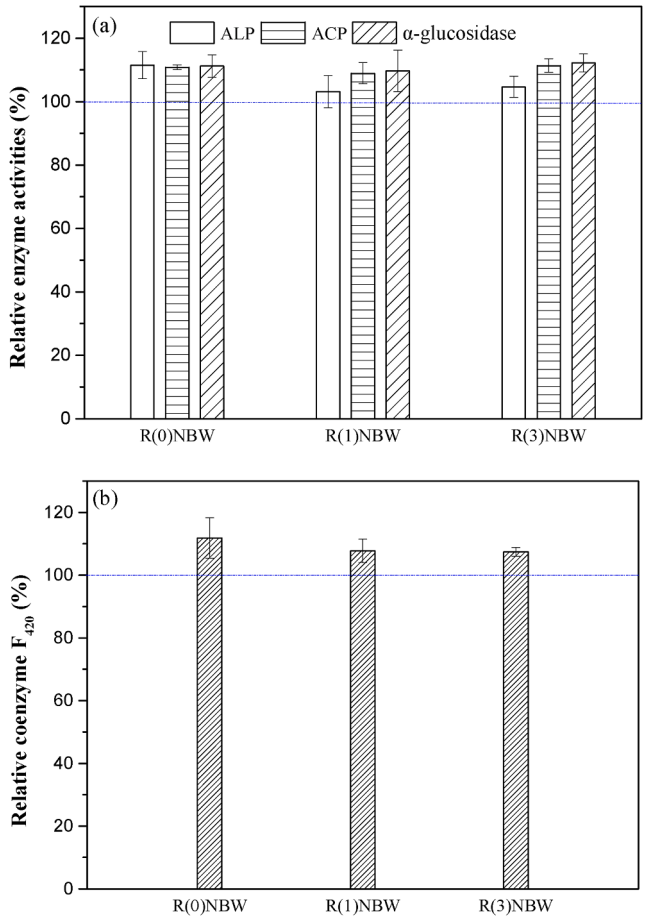


Fig. 5. Change of relative activities of extracellular hydrolases and coenzyme F_{420}; (a) Relative activities of extracellular hydrolases. (b) Relative activities of Coenzyme F_{420}.

Table 3
Main characteristics of NBW and DW used in this work.

Characteristics	Unit	Air-NBW	DW
NBs size	nm	160.00 ± 11.00	ND
NBs concentration	10^8 particles/mL	1.00 ± 0.10	ND
Zeta potential	mV	−17.30 ± 1.19	−1.76 ± 0.25
pH	–	5.85 ± 0.06	5.96 ± 0.03

ND (Not detectable).
Data in the table are expressed as Means ± SD.

methane production comes from the AM pathway, and the remaining 30% is contributed by the HM pathway (Yang et al., 2018). Although the proportion of methane production in the HM pathway is smaller than that of the AM pathway, the HM pathway can promote the mediation of biogasification under high ammonia conditions (Jiang et al., 2018). In addition, the HM pathway is the main pathway for methane production under ammonia inhibition (Yang et al., 2020b; Fotidis et al., 2013). Since the activity of HM was enhanced due to the supplementation of NBW, the mediating biogasification process may be promoted under ammonia inhibition. Therefore, despite the ammonia inhibition, the supplementation of NBW could promote methane production. The underlying mechanisms of NBW promoting the AD process and relieving ammonia inhibition still need further research.

This study indicated that NBW, as a novel method, can alleviate ammonia suppression and enhance methane yield. However, NBW could not change the poor methanogenesis performance when severe ammonia inhibition occurred (TAN > 7,000 mg/L). In addition, the increase in methane production under NBW supplementation was not high enough (from 12.3% to 38.7%) under ammonia inhibition. This was lower than the 54.6% achieved by using in-situ gas phase ammonia stripping (Yin et al., 2019), and was also lower than the 54.2% achieved by zero-valent iron addition (Yang et al., 2018). Although the increased amount in methane yield using the method of NBW supplementation was not high enough compared to other traditional methods, it enhances methane production with low-cost, environmentally friendly, and chemical-free. Moreover, the acquisition and application process of NBW does not require complicated equipment investment, huge energy consumption, or inconvenient operation techniques. With the continuous development of NBW technology, it is foreseeable that the production of NBW containing hundreds of millions or even billions of NBs concentrations will be realized. By then, NBW may not only greatly alleviate the ammonia inhibition, but also more effectively improve methane production compared to traditional methods. In addition, improving the application efficiency of NBW in the AD process, changing the application strategy of NBW, or combining NBW technology with optimized the parameters of AD process may also significantly increase the methane production under severe ammonia inhibition. Through the continuous innovation of NBW technology, the cost of applying NBW technology in the AD process to alleviate ammonia inhibition and increase methane production will be greatly reduced. With the cost reduction, NBW technology will be widely used in the actual AD industry. Moreover, underdeveloped areas will be able to solve the practical problems of the AD process by applying low-cost NBW technology. Therefore, the application of NBW to solve the ammonia inhibition that often occurs in the AD process of SM has excellent potential and is worthy of further study.

4. Conclusions

This work demonstrated the effect of NBW supplementation in alleviating ammonia inhibition and enhancing methane production in ammonia-rich SM. Results indicated that NBW supplementation facilitated the rapid increase of TA and PA and the reduction of VFA/TA during the early stage of AD system, thus improved buffering capacity and alleviated ammonia inhibition. Besides, the cumulative methane production was enhanced by 12.3–38.7% via NBW supplementation even under ammonia inhibition. Moreover, the activities of hydrolase and coenzyme F_{420} were enhanced due to the supplementation of NBW, indicating the augmentation of hydrolysis and methanogenesis in the AD system.

CRediT authorship contribution statement

Yujie Fan: Data curation, Formal analysis, Conceptualization, Writing - original draft, Methodology, Investigation, Validation. **Xiaojing Yang:** Investigation, Validation. **Zhongfang Lei:** Conceptualization, Writing - review & editing, Supervision. **Zhenya Zhang:** Conceptualization, Writing - review & editing, Supervision. **Motoyoshi Kobayashi:** Resources. **Yasuhisa Adachi:** Resources. **Kazuya Shimizu:** Conceptualization, Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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